

2 The Laws of Thermodynamics

According to the first law of thermodynamics, the internal energy of a system can be increased either by adding energy to the system or by doing work on it. That means the internal energy of a system, which is just the sum of the molecular kinetic and potential energies, can change as a result of two separate types of energy transfer across the boundary of the system. Although the first law imposes conservation of energy for both energy added by heat and work done on a system, it doesn't predict which of several possible energy-conserving processes actually occur in nature.

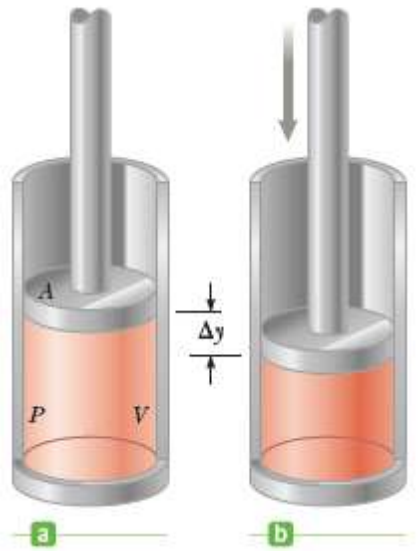
The second law of thermodynamics constrains the first law by establishing which processes allowed by the first law actually occur. For example, the second law tells us that energy never flows by heat spontaneously from a cold object to a hot object. One important application of this law is in the study of heat engines (such as the internal combustion engine) and the principles that limit their efficiency.

2.1 Work in Thermodynamics Processes

Energy can be transferred to a system by heat and by work done on the system. In most cases of interest treated here, the system is a volume of gas, which is important in understanding engines. All such systems of gas will be assumed to be in thermodynamic equilibrium, so that every part of the gas is at the same temperature and pressure. If that were not the case, the ideal gas law wouldn't apply and most of the results presented here wouldn't be valid. Consider a gas contained by a cylinder fitted with a movable piston and in equilibrium. The gas occupies a volume V and exerts a uniform pressure P on the cylinder walls and the piston. The gas is compressed slowly enough so the system remains

essentially in thermodynamic equilibrium at all times. As the piston is pushed downward by an external force F through a displacement Δy, the work done on the gas is

$$W = -F\Delta y = -PA\,\Delta y$$



where we have set the magnitude F of the external force equal to PA, possible because the pressure is the same everywhere in the system (by the assumption of equilibrium). Note that if the piston is pushed downward, $\Delta y = y_f - y_i$ is negative, so we need an explicit negative sign in the expression for W to make the work positive. The change in volume of the gas is $\Delta V = A\,\Delta y$, which leads to the following definition:

The **work W done on a gas** at constant pressure is given by

$$W = -P\Delta V \qquad \textbf{[2.1]}$$

where P is the pressure throughout the gas and DV is the change in volume of the gas during the process.

Figure 2.1 (a) A gas in a cylinder occupying a volume *V* at a pressure *P*. (b) Pushing the piston down compresses the gas.

If the gas is compressed as in Active Figure 2.1b, DV is negative and the work done on the gas is positive. If the gas expands, ΔV is positive and the work done on the gas is negative. The work done by the gas on its environment, W_{env}, is simply the negative of the work done on the gas. In the absence of a change in volume, the work is zero.

The definition of work W in Equation 2.1 specifies work done on a gas. In many texts, work W is defined as work done by a gas. In this text, work done by a gas is denoted by W_{env}. In every case, $W = -W_{env}$, so the two definitions differ by a minus sign. The reason it's important to define work W as work done on a gas is to make the concept of work in thermodynamics consistent with the concept of work in mechanics. In mechanics, the system is some object, and when positive work is done on that object, its energy increases. When work W done on a gas as defined in Equation 2.1 is positive, the internal energy of the gas increases, which is consistent with the mechanics definition. In Figure 2.2a the man pushes a crate, doing positive work on it, so the crate's speed and therefore its kinetic energy both increase. In Figure 2.2b a man pushes a piston to the right, compressing the gas in the container and doing positive work on the gas. The average speed of the molecules of gas increases, so the temperature and therefore the internal energy of the gas increase. Consequently, just as doing work on a crate increases its kinetic energy, doing work on a system of gas increases its internal energy.

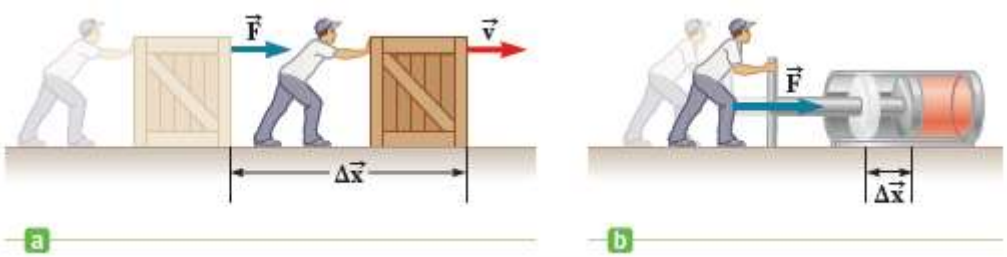


Figure 2.2 (a) When a force is exerted on a crate, the work done by that force increases the crate's mechanical energy. (b) When a piston is pushed, the gas in the container is compressed, increasing the gas's thermal energy.

PROBLEM In a system similar to that shown in Active Figure 2.1, the gas in the cylinder is at a pressure equal to 1.01×10^5 Pa and the piston has an area of 0.100 m^2. As energy is slowly added to the gas by heat, the piston is pushed up a distance of 4.00 cm. Calculate the work done by the expanding gas on the surroundings, W_{env}, assuming the pressure remains constant.

STRATEGY The work done on the environment is the negative of the work done on the gas given in Equation 2.1. Compute the change in volume and multiply by the pressure.

SOLUTION

Find the change in volume of the gas, ΔV, which is the cross-sectional area times the displacement:

$$\Delta V = A\,\Delta y = (0.100\ \text{m}^2)(4.00 \times 10^{-2}\ \text{m}) = 4.00 \times 10^{-3}\ \text{m}^3$$

Multiply this result by the pressure, getting the work the gas does on the environment, W_{env}:

$$W_{env} = P\,\Delta V = (1.01 \times 10^5\ \text{Pa})(4.00 \times 10^{-3}\ \text{m}^3) = 404\ \text{J}$$

REMARKS The volume of the gas increases, so the work done on the environment is positive. The work done on the system during this process is W = -404 J. The energy required to perform positive work on the environment must come from the energy of the gas.

Equation 2.1 can be used to calculate the work done on the system only when the pressure of the gas remains constant during the expansion or compression. A process in which the pressure remains constant is called an isobaric process. The pressure vs. volume graph, or PV diagram, of an isobaric process is shown in Figure 2.3. The curve on such a graph is called the *path* taken between the initial and final states, with the arrow indicating the direction the process is going, in this case from smaller to larger volume. The area under the graph is

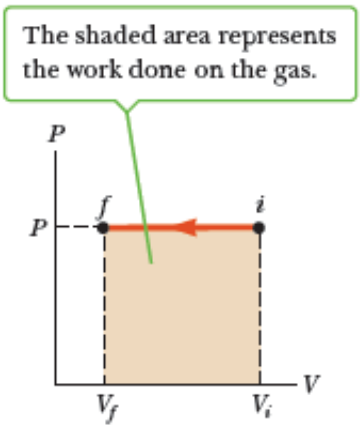


Figure 2.3 The *PV* diagram for a gas Being compressed at constant pressure.

$$\text{Area} = P(V_f - V_i) = P\,\Delta V$$

The area under the graph in a PV diagram is equal in magnitude to the work done on the gas.

That statement is true in general, whether or not the process proceeds at constant pressure. Just draw the PV diagram of the process, find the area underneath the graph (and above the horizontal axis), and that area will be the equal to the magnitude of the work done on the gas. If the arrow on the graph points toward larger volumes, the work done on the gas is negative. If the arrow on the graph points toward smaller volumes, the work done on the gas is positive.

Whenever negative work is done on a system, positive work is done by the system on its environment. The negative work done on the system represents a loss of energy from the system—the cost of doing positive work on the environment.

2.2 The First Law of Thermodynamics

The **first law of thermodynamics** is another energy conservation law that relates changes in internal energy—the energy associated with the position and jiggling of all the molecules of a system—to energy transfers due to heat and work. The first law is universally valid, applicable to all kinds of processes, providing a connection between the microscopic and macroscopic worlds.

There are two ways energy can be transferred between a system and its surrounding environment: by doing work, which requires a macroscopic displacement of an object through the application of a force, and by a direct exchange of energy across the system boundary, often by heat. Heat is the transfer of energy between a system and its environment due to a temperature difference and usually occurs through one or more of the mechanisms of radiation, conduction, and convection. For example, in Figure 2.5 hot gases and radiation impinge on the cylinder, raising its temperature, and energy Q is transferred by conduction to the gas, where it is distributed mainly through convection. Other processes for transferring energy into a system are possible, such as a chemical reaction or an electrical discharge. Any energy Q exchanged between the system and the environment and any work done through the expansion or compression of the system results in a change in the internal energy, DU, of the system. A change in internal energy results in measurable changes in the macroscopic variables of the system such as the pressure, temperature, and volume. The relationship between the change in internal energy, ΔU, energy Q, and the work W done on the system is given by the first law of thermodynamics:

> If a system undergoes a change from an initial state to a final state, then the change in the internal energy ΔU is given by
>
> $$\Delta U = U_f - U_i = Q + W \qquad [2.2]$$
>
> where Q is the energy exchanged between the system and the environment, and W is the work done on the system.

The quantity Q is positive when energy is transferred into the system and negative when energy is removed from the system. Figure 2.5 illustrates the first law for a cylinder of gas and how the system interacts with the environment. The gas cylinder contains a frictionless piston, and the block is initially at rest. Energy Q is introduced into the gas as the gas expands against the piston with constant pressure P. Until the piston hits the stops, it exerts a force on the block, which accelerates on a frictionless surface. Negative work W is done on the gas, and at the same time positive work $W_{env} = -W$ is done by the gas on the block. Adding the work done on the environment, W_{env}, to the work done on the gas, W, gives zero network, as it must because energy must be conserved.

From Equation 2.2 we also see that the internal energy of any isolated system must remain constant, so that $\Delta U = 0$. Even when a system isn't isolated, the change in internal energy will be zero if the system goes through a cyclic process in which all the thermodynamic variables—pressure, volume, temperature, and moles of gas—return to their original values.

It's important to remember that the quantities in Equation 2.2 concern a system, not the effect on the system's environment through work. If the system is hot gas expanding against a piston, as in Figure 2.5, the system work W is negative because the piston can only expand at the expense of the internal energy of the gas. The work W_{env} done by the hot gas on the environment—in this case, moving a piston which moves the block—is positive, but that's not the work W in Equation 2.2. This way of defining work in the first law makes it consistent with the concept of work defined in Chapter 5. In

both the mechanical and thermal cases, the effect on the system is the same: positive work increases the system's energy, and negative work decreases the system's energy.

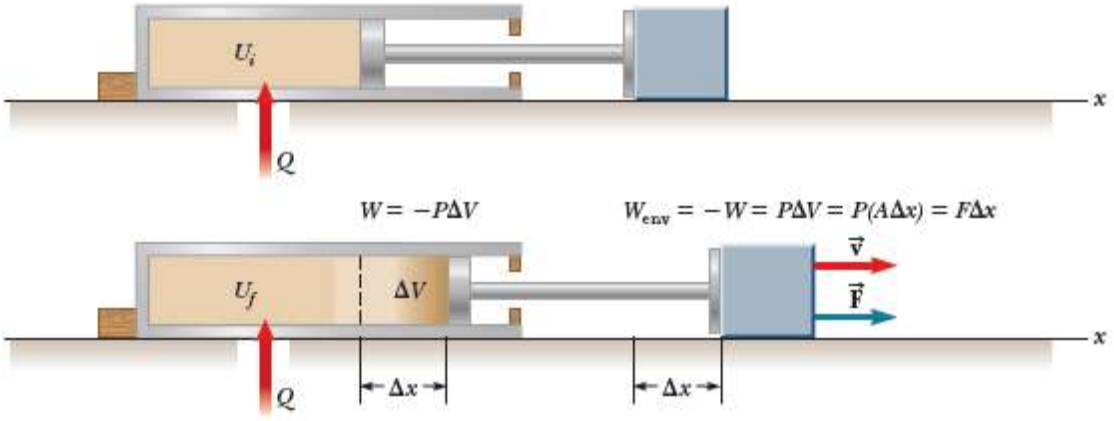


Figure 2.5 Thermal energy Q is transferred to the gas, increasing its internal energy. The gas presses against the piston, displacing it and performing mechanical work on the environment or, equivalently, doing negative work on the gas, reducing the internal energy.

Some textbooks identify W as the work done by the gas on its environment. This is an equivalent formulation, but it means that W must carry a minus sign in the first law. That convention isn't consistent with previous discussions of the energy of a system, because when W is positive the system loses energy, whereas in Chapter 5 positive W means that the system gains energy. For that reason, the old convention is not used in this book.

PROBLEM An ideal gas absorbs 5.00×10^3 J of energy while doing 2.00×10^3 J of work on the environment during a constant pressure process. (a) Compute the change in the internal energy of the gas. (b) If the internal energy now drops by 4.50×10^3 J and 7.50×10^3 J is expelled from the system, find the change in volume, assuming a constant pressure process at 1.01×10^5 Pa.

STRATEGY Part (a) requires substitution of the given information into the first law, Equation 2.2. Notice, however, that the given work is done on the environment. The negative of this amount is the work done on the system, representing a loss of internal energy. Part (b) is a matter of substituting the equation for work at constant pressure into the first law and solving for the change in volume.

SOLUTION

(a) Compute the change in internal energy of the gas.

Substitute values into the first law, noting that the work done on the gas is negative:

$$\Delta U = Q + W = 5.00 \times 10^3\,\text{J} - 2.00 \times 10^3\,\text{J}$$
$$= \boxed{3.00 \times 10^3\,\text{J}}$$

(b) Find the change in volume, noting that ΔU and Q are both negative in this case.

Substitute the equation for work done at constant pressure into the first law:

$$\Delta U = Q + W = Q - P\,\Delta V$$
$$-4.50 \times 10^3\,\text{J} = -7.50 \times 10^3\,\text{J} - (1.01 \times 10^5\,\text{Pa})\Delta V$$

Solve for the change in volume, ΔV:

$$\Delta V = \boxed{-2.97 \times 10^{-2}\,\text{m}^3}$$

REMARKS The change in volume is negative, so the system contracts, doing negative work on the environment, whereas the work W on the system is positive.

Recall that an expression for the internal energy of an ideal gas is

$$U = \tfrac{3}{2}nRT \qquad [2.3a]$$

This expression is valid only for a monatomic ideal gas, which means the particles of the gas consist of single atoms. The change in the internal energy, ΔU, for such a gas is given by

$$\Delta U = \tfrac{3}{2}nR\Delta T \qquad [2.3b]$$

The **molar specific heat at constant volume** of a monatomic ideal gas, Cv, is defined by

$$C_v \equiv \tfrac{3}{2}R \qquad [2.4]$$

The change in internal energy of an ideal gas can then be written

$$\Delta U = nC_v\,\Delta T \qquad [2.5]$$

For ideal gases, this expression is always valid, even when the volume isn't constant. The value of the molar specific heat, however, depends on the gas and can vary under different conditions of temperature and pressure.

Gas	Molar Specific Heat (J/mol · K)[a]			
	C_p	C_v	$C_p - C_v$	$\gamma = C_p/C_v$
Monatomic Gases				
He	20.8	12.5	8.33	1.67
Ar	20.8	12.5	8.33	1.67
Ne	20.8	12.7	8.12	1.64
Kr	20.8	12.3	8.49	1.69
Diatomic Gases				
H_2	28.8	20.4	8.33	1.41
N_2	29.1	20.8	8.33	1.40
O_2	29.4	21.1	8.33	1.40
CO	29.3	21.0	8.33	1.40
Cl_2	34.7	25.7	8.96	1.35
Polyatomic Gases				
CO_2	37.0	28.5	8.50	1.30
SO_2	40.4	31.4	9.00	1.29
H_2O	35.4	27.0	8.37	1.30
CH_4	35.5	27.1	8.41	1.31

[a]All values except that for water were obtained at 300 K.

Table 2.1 Molar Specific Heats of Various Gases

2.3 Thermal Processes

Engine cycles can be complex. Fortunately, they can often be broken down into a series of simple processes. In this section the four most common processes will be studied and illustrated by their effect on an ideal gas. Each process corresponds to making one of the variables in the ideal gas law a constant or assuming one of the three quantities in the first law of thermodynamics is zero. The four processes are called isobaric (constant pressure), adiabatic (no thermal energy transfer, or Q = 0), isovolumetric (constant volume, corresponding to W = 0) and isothermal (constant temperature, corresponding to $\Delta U = 0$). Naturally, many other processes don't fall into one of these four categories, so they will be covered in a fifth category, called a general process. What is essential in each case is to be able to calculate the three thermodynamic quantities in the first law: the work W, the thermal energy transfer Q, and the change in the internal energy ΔU.

Isobaric Processes

Recall from Section 2.1 that in an isobaric process the pressure remains constant as the gas expands or is compressed. An expanding gas does work on its environment, given by $W_{env} = P\,\Delta V$. The PV diagram of an isobaric expansion is given in Figure 2.3. As previously discussed, the magnitude of the work done on the gas is just the area under the path in its PV diagram: height times length, or $P\,\Delta V$. The negative of this quantity, $W = -P\,\Delta V$, is the energy lost by the gas because the gas does work as it expands. This is the quantity that should be substituted into the first law.

The work done by the gas on its environment must come at the expense of the change in its internal energy, ΔU. Because the change in the internal energy of an ideal gas is given by $\Delta U = nC_v\,\Delta T$, the temperature of an expanding gas must decrease as the internal energy decreases. Expanding volume and decreasing temperature means the pressure must also decrease, in conformity with the ideal gas law, PV = nRT. Consequently, the only way such a process can remain at constant pressure is if thermal energy Q is transferred into the gas by heat. Rearranging the first law, we obtain

$$Q = \Delta U - W = \Delta U + P\,\Delta V$$

Now we can substitute the expression in Equation 2.3b for ΔU and use the ideal gas law to substitute $P\,\Delta V = nR\,\Delta T$:

$$Q = \tfrac{3}{2}nR\,\Delta T + nR\,\Delta T = \tfrac{5}{2}nR\,\Delta T$$

Another way to express this transfer by heat is

$$Q = nC_p\,\Delta T \qquad [2.6]$$

where $C_p = \frac{5}{2}R$. For ideal gases, the molar heat capacity at constant pressure, C_p, is the sum of the molar heat capacity at constant volume, C_v, and the gas constant R:

$$C_p = C_v + R \qquad [2.7]$$

This can be seen in the fourth column of Table 2.1, where $C_p - C_v$ is calculated for a number of different gases. The difference works out to be approximately R in virtually every case.

Adiabatic Processes

In an adiabatic process, no energy enters or leaves the system by heat. Such a system is insulated, thermally isolated from its environment. In general, however, the system isn't mechanically isolated, so it can still do work. A sufficiently rapid process may be considered approximately adiabatic because there isn't time for any significant transfer of energy by heat.

For adiabatic processes Q = 0, so the first law becomes

$$\Delta U = W \quad \text{(adiabatic processes)}$$

The work done during an adiabatic process can be calculated by finding the change in the internal energy. Alternately, the work can be computed from a PV diagram. For an ideal gas undergoing an adiabatic process, it can be shown that

$$PV^{\gamma} = \text{constant} \qquad \textbf{[2.8a]}$$

Where

$$\gamma = \frac{C_p}{C_v} \qquad \textbf{[2.8a]}$$

is called the adiabatic index of the gas. Values of the adiabatic index for several different gases are given in Table 2.1. After computing the constant on the right-hand side of Equation 12.8a and solving for the pressure P, the area under the curve in the PV diagram can be found by counting boxes, yielding the work.

If a hot gas is allowed to expand so quickly that there is no time for energy to enter or leave the system by heat, the work done on the gas is negative and the internal energy decreases. This decrease occurs because kinetic energy is transferred from the gas molecules to the moving piston. Such an adiabatic expansion is of practical importance and is nearly realized in an internal combustion engine when a gasoline–air mixture is ignited and expands rapidly against a piston. The following example illustrates this process.

PROBLEM A monatomic ideal gas at an initial pressure of 1.01×10^5 Pa expands adiabatically from an initial volume of 1.50 m^3, doubling its volume (Figure 2.6). (a) Find the new pressure. (b) Sketch the PV diagram and estimate the work done on the gas.

STRATEGY There isn't enough information to solve this problem with the ideal gas law. Instead, use Equation 2.8a, b and the given information to find the adiabatic index and the constant C for the process. For part (b), sketch the PV diagram and count boxes to estimate the area under the graph, which gives the work.

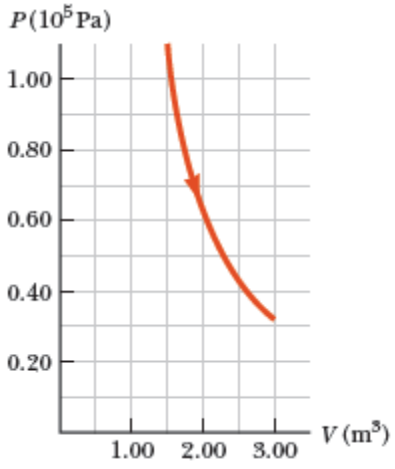


Figure 12.6 (Example 2.6) The *PV* diagram of an adiabatic expansion: the graph of $P = CV^{-\gamma}$, where *C* is a constant and g 5 C_p/C_v.

SOLUTION

(a) Find the new pressure.

First, calculate the adiabatic index:

$$\gamma = \frac{C_p}{C_v} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}$$

Use Equation 12.8a to find the constant C:

$$C = P_1V_1^{\gamma} = (1.01 \times 10^5 \text{ Pa})(1.50 \text{ m}^3)^{5/3}$$
$$= 1.99 \times 10^5 \text{ Pa} \cdot \text{m}^5$$

The constant C is fixed for the entire process and can be used to find P_2:

$$C = P_2V_2^{\gamma} = P_2(3.00 \text{ m}^3)^{5/3}$$
$$1.99 \times 10^5 \text{ Pa} \cdot \text{m}^5 = P_2\,(6.24 \text{ m}^5)$$
$$P_2 = 3.19 \times 10^4 \text{ Pa}$$

(b) Estimate the work done on the gas from a PV diagram.

Count the boxes between $V_1 = 1.50 \text{ m}^3$ and $V_2 = 3.00 \text{ m}^3$ in the graph of $P = (1.99 \times 10^5 \text{ Pa} \cdot \text{m}^5)V^{-5/3}$ in the PV diagram shown in Figure 12.6:

Number of boxes ≈ 17

Each box has an 'area' of 5.00×10^3 J.

$$W \approx -17 \cdot 5.00 \times 10^3 \text{ J} = -8.5 \times 10^4 \text{ J}$$

REMARKS The exact answer, obtained with calculus, is -8.43 × 10^4 J, so our result is a very good estimate. The answer is negative because the gas is expanding, doing positive work on the environment, thereby reducing its own internal energy.

Isovolumetric Processes

An isovolumetric process, sometimes called an isochoric process (which is harder to remember), proceeds at constant volume, corresponding to vertical lines in a PV diagram. If the volume doesn't change, no work is done on or by the system, so W = 0 and the first law of thermodynamics reads

$$\Delta U = Q \quad \text{(isovolumetric process)}$$

This result tells us that in an isovolumetric process, the change in internal energy of a system equals the energy transferred to the system by heat. From Equation 2.5, the energy transferred by heat in constant volume processes is given by

$$Q = nC_v\,\Delta T \qquad [2.9]$$

PROBLEM A monatomic ideal gas has a temperature T 5 3.00 3 102 K and a constant volume of 1.50 L. If there are 5.00 moles of gas, (a) how much thermal energy must be added in order to raise the temperature of the gas to 3.80 3 102 K? (b) Calculate the change in pressure of the gas, DP. (c) How much thermal energy would be required if the gas were ideal and diatomic? (d) Calculate the change in the pressure for the diatomic gas.

REMARKS The constant volume diatomic gas, under the same conditions, requires more thermal energy per degree of temperature change because there are more ways for the diatomic molecules to store energy. Despite the extra energy added, the diatomic gas reaches the same final pressure as the monatomic gas.

SOLUTION

(a) How much thermal energy must be added in order to raise the temperature of the gas to 3.80×10^2 K?

Apply Equation 12.9, using the fact that $C_v = 3R/2$ for an ideal monatomic gas:

$$(1) \quad Q = \Delta U = nC_v\Delta T = \tfrac{3}{2}nR\Delta T$$
$$= \tfrac{3}{2}(5.00\ \text{mol})(8.31\ \text{J/K}\cdot\text{mol})(80.0\ \text{K})$$
$$Q = \boxed{4.99 \times 10^3\ \text{J}}$$

(b) Calculate the change in pressure, ΔP.

Use the ideal gas equation $PV = nRT$ and Equation (1) to relate ΔP to Q:

$$\Delta(PV) = (\Delta P)V = nR\Delta T = \tfrac{2}{3}Q$$

Solve for ΔP:

$$\Delta P = \frac{2}{3}\frac{Q}{V} = \frac{2}{3}\frac{4.99 \times 10^3\ \text{J}}{1.50 \times 10^{-3}\ \text{m}^3}$$
$$= \boxed{2.22 \times 10^6\ \text{Pa}}$$

(c) How much thermal energy would be required if the gas were ideal and diatomic?

Repeat the calculation with $C_v = 5R/2$:

$$Q = \Delta U = nC_v\Delta T = \frac{5}{2}nR\Delta T = 8.31 \times 10^3\ \text{J}$$

(d) Calculate the change in the pressure for the diatomic gas.

Use the result of part (c) and repeat the calculation of part (b), with 2/3 replaced by 2/5 because the gas is diatomic:

$$\Delta P = \frac{2}{5}\frac{Q}{V} = \frac{2}{5}\frac{8.31 \times 10^3\ \text{J}}{1.50 \times 10^{-3}\ \text{m}^3}$$
$$= \boxed{2.22 \times 10^6\ \text{Pa}}$$

Isothermal Processes

During an isothermal process, the temperature of a system doesn't change. In an ideal gas the internal energy U depends only on the temperature, so it follows that $\Delta U = 0$ because $\Delta T = 0$. In this case, the first law of thermodynamics gives

$$W = -Q \quad \text{(isothermal process)}$$

We see that if the system is an ideal gas undergoing an isothermal process, the work done on the system is equal to the negative of the thermal energy transferred to the system. Such a process can be visualized in Figure 2.7. A cylinder filled with gas is in contact with a large energy reservoir that can exchange energy with the gas without changing its temperature. For a constant temperature ideal gas,

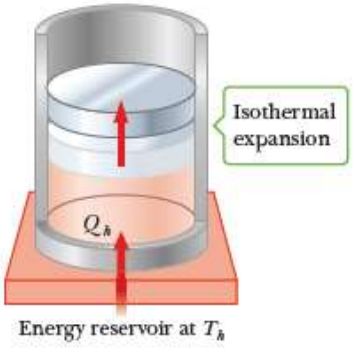


Figure 2.7 The gas in the cylinder expands isothermally

$$P = \frac{nRT}{V}$$

where the numerator on the right-hand side is constant. The PV diagram of a typical isothermal process is graphed in Figure 2.8, contrasted with an adiabatic process. The pressure falls off more rapidly for an adiabatic expansion because thermal energy can't be transferred into the system. In an isothermal expansion, the system loses energy by doing work on the environment but regains an equal amount of energy across the boundary. Using methods of calculus, it can be shown that the work done on the environment during an isothermal process is given by

$$W_{\text{env}} = nRT \ln\left(\frac{V_f}{V_i}\right)$$

The symbol "ln" in Equation 2.10 is an abbreviation for the natural logarithm, discussed in Appendix A. The work W done on the gas is just the negative of W_{env}.

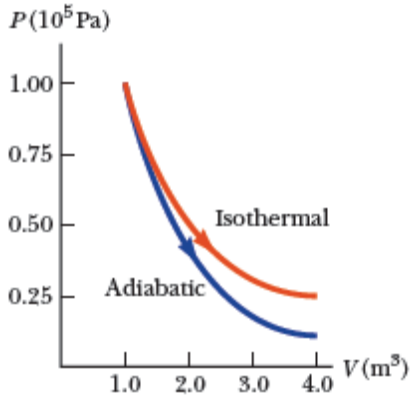


Figure 2.8 The *PV* diagram of an isothermal expansion,graph

PROBLEM A balloon contains 5.00 moles of a monatomic ideal gas. As energy is added to the system by heat (say, by absorption from the Sun), the volume increases by 25% at a constant temperature of 27.0°C. Find the work W_{env} done by the gas in expanding the balloon, the thermal energy Q transferred to the gas, and the work W done on the gas.

STRATEGY Be sure to convert temperatures to kelvins. Use Equation 12.10 for isothermal work W_{env} done on the environment to find the work W done on the balloon, which satisfies $W = -W_{env}$. Further, for an isothermal process, the thermal energy Q transferred to the system equals the work W_{env} done by the system on the environment.

SOLUTION

Substitute into Equation 12.10, finding the work done during the isothermal expansion. Note that $T = 27.0°C = 3.00 \times 10^2$ K.

$$W_{env} = nRT \ln\left(\frac{V_f}{V_i}\right)$$
$$= (5.00 \text{ mol})(8.31 \text{ J/K} \cdot \text{mol})(3.00 \times 10^2 \text{ K}) \times \ln\left(\frac{1.25V_0}{V_0}\right)$$
$$W_{env} = 2.78 \times 10^3 \text{ J}$$
$$Q = W_{env} = 2.78 \times 10^3 \text{ J}$$

The negative of this amount is the work done on the gas:

$$W = -W_{env} = -2.78 \times 10^3 \text{ J}$$

REMARKS Notice the relationship between the work done on the gas, the work done on the environment, and the energy transferred. These relationships are true of all isothermal processes

General Case

When a process follows none of the four given models, it's still possible to use the first law to get information about it. The work can be computed from the area under the curve of the PV diagram, and if the temperatures at the endpoints can be found, ΔU follows from Equation 2.5, as illustrated in the following example.

PROBLEM A quantity of 4.00 moles of a monatomic ideal gas expands from an initial volume of 0.100 m^3 to a final volume of 0.300 m^3 and pressure of 2.5×10^5 Pa (Fig. 2.9a). Compute (a) the work done

on the gas, (b) the change in internal energy of the gas, and (c) the thermal energy transferred to the gas.

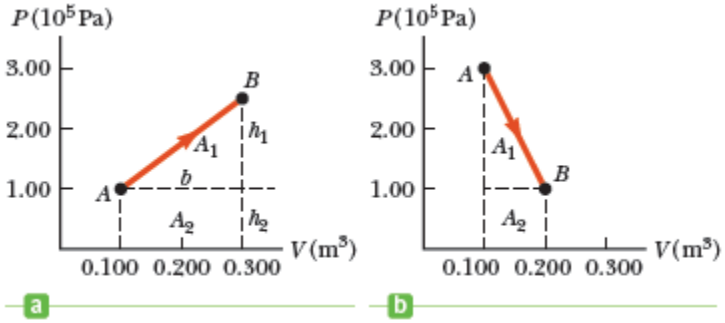


Figure 2.9 (a) (Example 12.9) (b) (Exercise 12.9)

STRATEGY The work done on the gas is equal to the negative of the area under the curve in the PV diagram. Use the ideal gas law to get the temperature change and, subsequently, the change in internal energy. Finally, the first law gives the thermal energy transferred by heat.

SOLUTION

(a) Find the work done on the gas by computing the area under the curve in Figure 12.9a.

Find A_1, the area of the triangle:

$$A_1 = \tfrac{1}{2}bh_1 = \tfrac{1}{2}(0.200\ \text{m}^3)(1.50 \times 10^5\ \text{Pa}) = 1.50 \times 10^4\ \text{J}$$

Find A_2, the area of the rectangle:

$$A_2 = bh_2 = (0.200\ \text{m}^3)(1.00 \times 10^5\ \text{Pa}) = 2.00 \times 10^4\ \text{J}$$

Sum the two areas (the gas is expanding, so the work done on the gas is negative and a minus sign must be supplied):

$$W = -(A_1 + A_2) = -3.50 \times 10^4\ \text{J}$$

(b) Find the change in the internal energy during the process.

Compute the temperature at points A and B with the ideal gas law:

$$T_A = \frac{P_A V_A}{nR} = \frac{(1.00 \times 10^5\ \text{Pa})(0.100\ \text{m}^3)}{(4.00\ \text{mol})(8.31\ \text{J/K}\cdot\text{mol})} = 301\ \text{K}$$

$$T_B = \frac{P_B V_B}{nR} = \frac{(2.50 \times 10^5\ \text{Pa})(0.300\ \text{m}^3)}{(4.00\ \text{mol})(8.31\ \text{J/K}\cdot\text{mol})} = 2.26 \times 10^3\ \text{K}$$

Compute the change in internal energy:

$$\Delta U = \tfrac{3}{2}nR\,\Delta T$$
$$= \tfrac{3}{2}(4.00\ \text{mol})(8.31\ \text{J/K}\cdot\text{mol})(2.26 \times 10^3\ \text{K} - 301\ \text{K})$$
$$\Delta U = 9.77 \times 10^4\ \text{J}$$

(c) Compute Q with the first law:

$$Q = \Delta U - W = 9.77 \times 10^4\ \text{J} - (-3.50 \times 10^4\ \text{J})$$
$$= 1.33 \times 10^5\ \text{J}$$

REMARKS As long as it's possible to compute the work, cycles involving these more exotic processes can be completely analyzed. Usually, however, it's necessary to use calculus. Note that the solution to part (b) could have been facilitated by yet another application of PV = nRT

$$\Delta U = \tfrac{3}{2}nR\Delta T = \tfrac{3}{2}\Delta(PV) = \tfrac{3}{2}(P_B V_B - P_A V_A)$$

This result means that even in the absence of information about the number of moles or the temperatures, the problem could be solved knowing the initial and final pressures and volumes.

Given all the different processes and formulas, it's easy to become confused when approaching one of these ideal gas problems, although most of the time only substitution into the correct formula is required. The essential facts and formulas are compiled in Table 2.2, both for easy reference and also to display the similarities and differences between the processes.

Table 2.2 The First Law and Thermodynamic Processes (Ideal Gases)

Process	ΔU	Q	W
Isobaric	$nC_v\Delta T$	$nC_p\Delta T$	$-P\Delta V$
Adiabatic	$nC_v\Delta T$	0	ΔU
Isovolumetric	$nC_v\Delta T$	ΔU	0
Isothermal	0	$-W$	$-nRT\ln\left(\frac{V_f}{V_i}\right)$
General	$nC_v\Delta T$	$\Delta U - W$	(PV Area)

2.4 Heat Engines and the Second Law of Thermodynamics

A **heat engine** takes in energy by heat and partially converts it to other forms, such as electrical and mechanical energy. In a typical process for producing electricity in a power plant, for instance, coal or some other fuel is burned, and the resulting internal energy is used to convert water to steam. The steam is then directed at the blades of a turbine, setting it rotating. Finally, the mechanical energy associated with this rotation is used to drive an electric generator. In another heat engine— the internal combustion engine in an automobile—energy enters the engine as fuel is injected into the cylinder and combusted, and a fraction of this energy is converted to mechanical energy.

In general, a heat engine carries some working substance through a **cyclic process** during which (1) energy is transferred by heat from a source at a high temperature, (2) work is done by the engine, and (3) energy is expelled by the engine by heat to a source at lower temperature. As an example, consider the operation of a steam engine in which the working substance is water. The water in the engine is carried through a cycle in which it first evaporates into steam in a boiler and
then expands against a piston. After the steam is condensed with cooling water, it returns to the boiler, and the process is repeated.

It's useful to draw a heat engine schematically, as in Active Figure 2.11. The engine absorbs energy Q_h from the hot reservoir, does work W_{eng}, then gives up energy Q_c to the cold reservoir. (Note that negative work is done on the engine, so that $W = -W_{eng}$.) Because the working substance goes through a cycle, always returning to its initial thermodynamic state, its initial and final internal energies are equal, so $\Delta U = 0$. From the first law of thermodynamics, therefore,

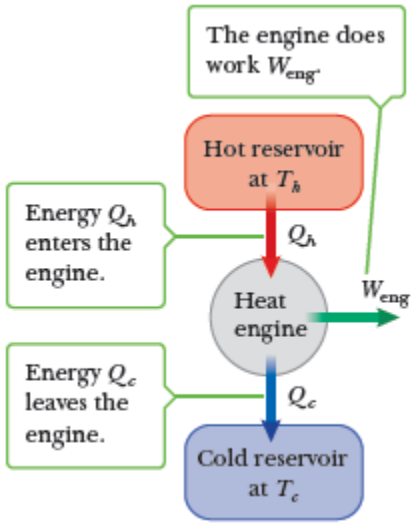
$$\Delta U = 0 = Q + W \rightarrow Q_{net} = -W = W_{eng}$$

The last equation shows that the work Weng done by a heat engine equals the net energy absorbed by the engine. As we can see from Active Figure 2.11, $Q_{net} = |Q_h| - |Q_c|$. Therefore,

$$W_{eng} = |Q_h| - |Q_c| \qquad [2.10]$$

Ordinarily, a transfer of thermal energy Q can be either positive or negative, so the use of absolute value signs makes the signs of Q_h and Q_c explicit.

If the working substance is a gas, then the work done by the engine for a cyclic process is the area enclosed by the curve representing the process on a PV diagram. This area is shown for an arbitrary cyclic process in Figure 2.12.



Active Figure 2.11 In this schematic representation of a heat engine, part of the thermal energy from the hot reservoir is turned into work while the rest is expelled to the cold reservoir.

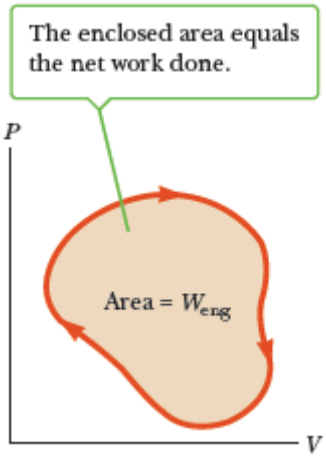


Figure 2.12 The *PV* diagram for an arbitrary cyclic process.

The **thermal efficiency** e of a heat engine is defined as the work done by the engine, W_{eng}, divided by the energy absorbed during one cycle:

$$e \equiv \frac{W_{eng}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|} \qquad [2.11]$$

We can think of thermal efficiency as the ratio of the benefit received (work) to the cost incurred (energy transfer at the higher temperature). Equation 2.11 shows that a heat engine has 100% efficiency (e = 1) only if $Q_c = 0$, meaning no energy is expelled to the cold reservoir. In other words, a heat engine with perfect efficiency would have to use all the input energy for doing mechanical work. That isn't possible.

PROBLEM During one cycle, an engine extracts 2.00×10^3 J of energy from a hot reservoir and transfers 1.50×10^3 J to a cold reservoir. (a) Find the thermal efficiency of the engine. (b) How much work does this engine do in one cycle? (c) What average power does the engine generate if it goes through four cycles in 2.50 s?

STRATEGY Apply Equation 2.11 to obtain the thermal efficiency, then use the first law, adapted to engines (Eq. 2.10), to find the work done in one cycle. To obtain the power generated, divide the work done in four cycles by the time it takes to run those cycles.

REMARKS Problems like this usually reduce to solving two equations and two unknowns, as here, where the two equations are the efficiency equation and the first law and the unknowns are the efficiency and the work done by the engine.

SOLUTION

(a) Find the engine's thermal efficiency.

Substitute Q_c and Q_h into Equation 12.12:

$$e = 1 - \frac{|Q_c|}{|Q_h|} = 1 - \frac{1.50 \times 10^3\,\text{J}}{2.00 \times 10^3\,\text{J}} = 0.250, \text{ or } 25.0\%$$

(b) How much work does this engine do in one cycle?

Apply the first law in the form of Equation 12.11 to find the work done by the engine:

$$W_{\text{eng}} = |Q_h| - |Q_c| = 2.00 \times 10^3\,\text{J} - 1.50 \times 10^3\,\text{J} = 5.00 \times 10^2\,\text{J}$$

(c) Find the average power output of the engine.

Multiply the answer of part (b) by four and divide by time:

$$P = \frac{W}{\Delta t} = \frac{4.00 \times (5.00 \times 10^2\,\text{J})}{2.50\,\text{s}} = 8.00 \times 10^2\,\text{W}$$

The Second Law of Thermodynamics

There are limits to the efficiency of heat engines. The ideal engine would convert all input energy into useful work, but it turns out that such an engine is impossible to construct. The Kelvin-Planck formulation **of the second law of thermodynamics** can be stated as follows:

> No heat engine operating in a cycle can absorb energy from a reservoir
> And use it entirely for the performance of an equal amount of work.

This form of the second law means that the efficiency $e = W_{eng}/|Q_h|$ of engines must always be less than 1. Some energy $|Q_c|$ must always be lost to the environment. In other words, it's theoretically impossible to construct a heat engine with an efficiency of 100%.

To summarize, the first law says **we can't get a greater amount of energy out of a cyclic process than we put in**, and the second law says **we can't break even**. No matter what engine is used, some energy must be transferred by heat to the cold reservoir. In Equation 2.10, the second law simply means $|Q_c|$ is always greater than zero.

There is another equivalent statement of the second law:

> If two systems are in thermal contact, net thermal energy transfers
> spontaneously by heat from the hotter system to the colder system.

Here, spontaneous means the energy transfer occurs naturally, with no work being done. Thermal energy naturally transfers from hotter systems to colder systems. Work must be done to transfer thermal energy from a colder system to a hotter system, however. An example is the refrigerator, which transfers thermal energy from inside the refrigerator to the warmer kitchen.

Reversible and Irreversible Processes

No engine can operate with 100% efficiency, but different designs yield different efficiencies, and it turns out that one design in particular delivers the maximum possible efficiency. This design is the Carnot cycle, discussed in the next subsection. Understanding it requires the concepts of reversible and irreversible processes. In a **reversible** process, every state along the path is an equilibrium state, so the system can return to its initial conditions by going along the same path in the reverse direction. A process that doesn't satisfy this requirement is **irreversible**.

Most natural processes are known to be irreversible; the reversible process is an idealization. Although real processes are always irreversible, some are almost reversible. If a real process occurs so slowly that the system is virtually always in equilibrium, the process can be considered reversible. Imagine compressing a gas very slowly by dropping grains of sand onto a frictionless piston, as in Figure 2.16. The temperature can be kept constant by placing the gas in thermal contact with an energy reservoir. The pressure, volume, and temperature of the gas are well defined during this isothermal compression.

Each added grain of sand represents a change to a new equilibrium state. The process can be reversed by slowly removing grains of sand from the piston

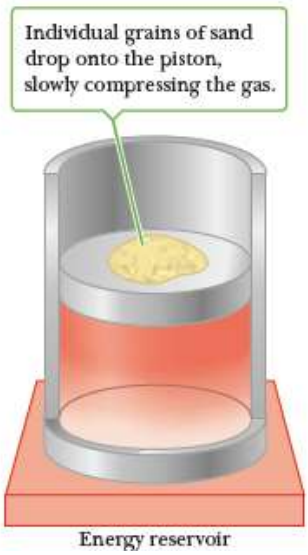


Figure 2.16 A method for compressing a gas in a reversible isothermal process.

The Carnot Engine

In 1824, in an effort to understand the efficiency of real engines, a French engineer named Sadi Carnot (1796–1832) described a theoretical engine now called a Carnot engine that is of great importance from both a practical and a theoretical viewpoint. He showed that a heat engine operating in an ideal, reversible cycle— now called a Carnot cycle—between two energy reservoirs is the most efficient engine possible. Such an engine establishes an upper limit on the efficiencies of all real engines. Carnot's theorem can be stated as follows:

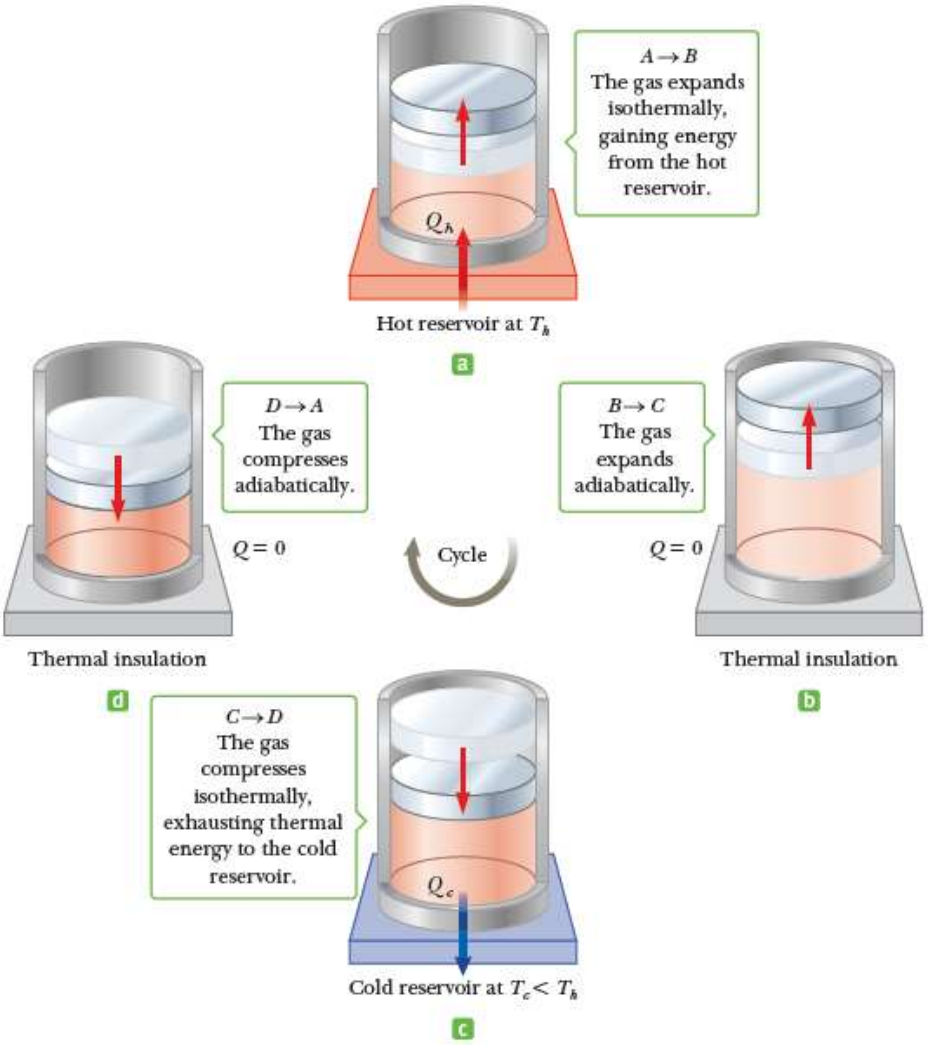
> No real engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs.

In a Carnot cycle, an ideal gas is contained in a cylinder with a movable piston at one end. The temperature of the gas varies between T_c and T_h. The cylinder walls and the piston are thermally nonconducting. Active Figure 2.17 shows the four stages of the Carnot cycle, and Active Figure 2.18 is the PV diagram for the cycle. The cycle consists of two adiabatic and two isothermal processes, all reversible:

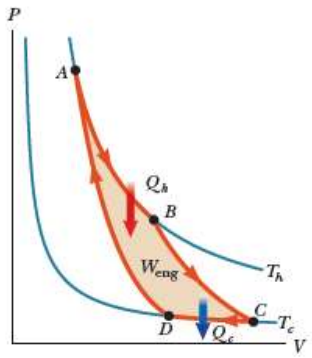
1. The process A $\rightarrow$ B is an isothermal expansion at temperature T_h in which the gas is placed in thermal contact with a hot reservoir (a large oven, for example) at temperature T_h (Active Fig. 2.17a). During the process, the gas absorbs energy Q_h from the reservoir and does work W_{AB} in raising the piston.
2. In the process B$\rightarrow$C, the base of the cylinder is replaced by a thermally nonconducting wall and the gas expands adiabatically, so no energy enters or leaves the system by heat (Active Fig. 2.17b). During the process, the temperature falls from T_h to T_c and the gas does work WBC in raising the piston.
3. In the process C$\rightarrow$D, the gas is placed in thermal contact with a cold reservoir at temperature T_c (Active Fig. 2.17c) and is compressed isothermally at temperature T_c. During this time, the gas expels energy Q_c to the reservoir and the work done on the gas is W_{CD}.

4. In the final process, D→A, the base of the cylinder is again replaced by a thermally nonconducting wall (Active Fig. 2.17d), and the gas is compressed adiabatically. The temperature of the gas increases to T_h, and the work done on the gas is W_{DA}.

For a Carnot engine, the following relationship between the thermal energy transfers and the absolute temperatures can be derived:



Active Figure 2.17 The Carnot cycle. The letters A, B, C, and D refer to the states of the gas shown in Active Figure 12.18. The arrows on the piston indicate the direction of its motion during each process.



Active Figure 2.18 The PV diagram for the Carnot cycle. The net work done, Weng,equals the net energy transferred into the Carnot engine in one cycle, |Qh| - |Qc|.

$$\frac{|Q_c|}{|Q_h|} = \frac{T_c}{T_h} \qquad [2.12]$$

Substituting this expression into Equation 2.11, we find that the thermal efficiency of a Carnot engine is

$$e_C = 1 - \frac{T_c}{T_h} \qquad [2.13]$$

where T must be in kelvins. From this result, we see that **all Carnot engines operating reversibly between the same two temperatures have the same efficiency**. Equation 2.13 can be applied to any working substance operating in a Carnot cycle between two energy reservoirs. According to that equation, the efficiency is zero if $T_c = T_h$. The efficiency increases as T_c is lowered and as T_h is increased. The efficiency can be one (100%), however, only if $T_c = 0$ K. According to the third law of thermodynamics, it's impossible to lower the temperature of a system to absolute zero in a finite number of steps, so such reservoirs are not available and the maximum efficiency is always less than 1. In most practical cases, the cold reservoir is near room temperature, about 300 K, so increasing the efficiency requires raising the temperature of the hot reservoir. **All real engines operate irreversibly, due to friction and the brevity of their cycles, and are therefore less efficient than the Carnot engine**.

PROBLEM A steam engine has a boiler that operates at 5.00×10^2 K. The energy from the boiler changes water to steam, which drives the piston. The temperature of the exhaust is that of the outside air, 3.00×10^2 K. (a) What is the engine's efficiency if it's an ideal engine? (b) If the 3.50×10^3 J of energy is supplied from the boiler, find the energy transferred to the cold reservoir and the work done by the engine on its environment.

STRATEGY This problem requires substitution into Equations 2.12 and 2.13, both applicable to a Carnot engine. The first equation relates the ratio Q_c/Q_h to the ratio T_c/T_h, and the second gives the Carnot engine efficiency.

SOLUTION

(a) Find the engine's efficiency, assuming it's ideal.

Substitute into Equation 12.16, the equation for the efficiency of a Carnot engine:

$$e_C = 1 - \frac{T_c}{T_h} = 1 - \frac{3.00 \times 10^2\ \text{K}}{5.00 \times 10^2\ \text{K}} = 0.400, \text{ or } 40.0\%$$

(b) Find the energy transferred to the cold reservoir and the work done on the environment if 3.50×10^3 J is delivered to the engine during one cycle.

Equation 12.15 shows that the ratio of energies equals the ratio of temperatures:

$$\frac{|Q_c|}{|Q_h|} = \frac{T_c}{T_h} \quad \rightarrow \quad |Q_c| = |Q_h| \frac{T_c}{T_h}$$

Substitute, finding the energy transferred to the cold reservoir:

$$|Q_c| = (3.50 \times 10^3\ \text{J})\left(\frac{3.00 \times 10^2\ \text{K}}{5.00 \times 10^2\ \text{K}}\right) = 2.10 \times 10^3\ \text{J}$$

Use Equation 12.11 to find the work done by the engine:

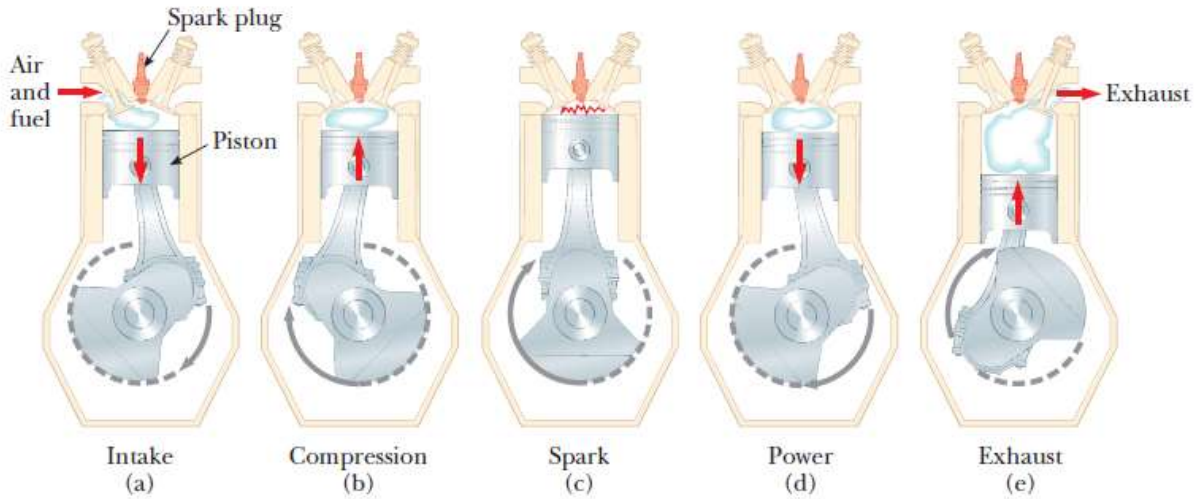
$$W_{\text{eng}} = |Q_h| - |Q_c| = 3.50 \times 10^3\ \text{J} - 2.10 \times 10^3\ \text{J} = 1.40 \times 10^3\ \text{J}$$

REMARKS This problem differs from the earlier examples on work and efficiency because we used the special Carnot relationships, Equations 2.12 and 2.13. Remember that these equations can only be used when the cycle is identified as ideal or a Carnot.

2.5 Gasoline and Diesel Engines

In a gasoline engine, six processes occur in each cycle; five of them are illustrated in Active Figure 2.19. In this discussion, let's consider the interior of the cylinder above the piston to be the system that is taken through repeated cycles in the engine's operation. For a given cycle, the piston moves up and down twice, which represents a four-stroke cycle consisting of two upstrokes and two downstrokes. The processes in the cycle can be approximated by the Otto cycle shown in the PV diagram in Active Figure 2.20. In the following discussion, refer to Active Figure 2.19 for the pictorial representation of the strokes and Active Figure 2.20 for the significance on the PV diagram of the letter designations below:

1. During the *intake stroke* O → A (Active Fig. 2.19a), the piston moves downward and a gaseous mixture of air and fuel is drawn into the cylinder at atmospheric pressure. In this process, the volume increases from V_2 to V_1. That is the energy input part of the cycle: energy enters the system (the interior of the cylinder) by matter transfer as potential energy stored in the fuel.
2. During the *compression stroke* A → B (Active Fig. 2.19b), the piston moves upward, the air-fuel mixture is compressed adiabatically from volume V*1* to volume V*2*, and the temperature increases from T_A to T_B. The work done on the gas is positive, and its value is equal to the negative of the area under the curve *AB* in Active Figure 2.20.



ACTIVE FIGURE 2.19

3. In process B → C, combustion occurs when the spark plug fires (Active Fig. 2.19c). That is not one of the strokes of the cycle because it occurs in a very short time interval while the piston is at its highest position. The combustion represents a rapid transformation from potential energy stored in chemical bonds in the fuel to internal energy associated with molecular motion, which is related to temperature. During this time interval, the mixture's pressure and temperature increase rapidly, with the temperature rising from TB to TC. The volume, however, remains approximately constant because of the short time interval. As a result, approximately no work is done on or by the gas. We can model this process in the PV diagram (Active Fig. 2.20) as that process in which the energy $|Q_h|$ enters the

system. (In reality, however, this process is a conversion of energy already in the cylinder from process O → A.)

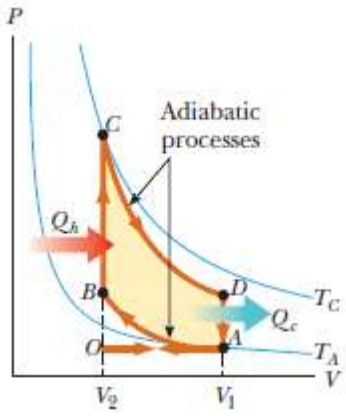
4. In the *power stroke* C → D (Active Fig. 2.19d), the gas expands adiabatically from V_2 to V_1. This expansion causes the temperature to drop from T_C to T_D. Work is done by the gas in pushing the piston downward, and the value of this work is equal to the area under the curve CD.
5. In the process D → A (not shown in Active Fig. 2.19), an exhaust valve is opened as the piston reaches the bottom of its travel and the pressure suddenly drops for a short time interval. During this time interval, the piston is almost stationary and the volume is approximately constant. Energy is expelled from the interior of the cylinder and continues to be expelled during the next process.
6. In the final process, *the exhaust stroke* A → O (Active Fig. 2.19e), the piston moves upward while the exhaust valve remains open. Residual gases are exhausted at atmospheric pressure, and the volume decreases from V_1 to V_2. The cycle then repeats.

If the air–fuel mixture is assumed to be an ideal gas, the efficiency of the Otto cycle is

$$e = 1 - \frac{1}{(V_1/V_2)^{\gamma-1}} \quad \text{(Otto cycle)} \qquad [2.14]$$

where V_1/V_2 is the compression ratio and γ is the ratio of the molar specific heats C_P/C_V for the fuel–air mixture. Equation 2.14, shows that the efficiency increases as the compression ratio increases. For a typical compression ratio of 8 and with $\gamma = 1.4$, Equation 2.14 predicts a theoretical efficiency of 56% for an engine operating in the idealized Otto cycle. This value is much greater than that achieved in real engines (15% to 20%) because of such effects as friction, energy transfer by conduction through the cylinder walls, and incomplete combustion of the air–fuel mixture.

Diesel engines operate on a cycle similar to the Otto cycle, but they do not employ a spark plug. The compression ratio for a diesel engine is much greater than that for a gasoline engine. Air in the cylinder is compressed to a very small volume, and, as a consequence, the cylinder temperature at the end of the compression stroke is very high. At this point, fuel is injected into the cylinder. The temperature is high enough for the fuel–air mixture to ignite without the assistance of a spark plug. Diesel engines are more efficient than gasoline engines because of their greater compression ratios and resulting higher combustion temperatures.



ACTIVE FIGURE 2.20 *PV* diagram for the Otto cycle, which approximately represents the processes occurring in an internal combustion engine.

2.6 Entropy

Temperature and internal energy, associated with the zeroth and first laws of thermodynamics respectively, are both state variables, meaning they can be used to describe the thermodynamic state of a system. A state variable called the entropy S is related to the second law of thermodynamics. We define entropy on a macroscopic scale as German physicist Rudolf Clausius (1822–1888) first expressed it in
1865:

> Let Q_r be the energy absorbed or expelled during a reversible, constant, temperature process between two equilibrium states. Then the change in entropy during any constant temperature process connecting the two equilibrium states is defined as
>
> $$\Delta S = \frac{Q_r}{T} \qquad [2.14]$$
>
> **SI unit: joules/kelvin (J/K)**

A similar formula holds when the temperature isn't constant, but its derivation entails calculus and won't be considered here. Calculating the change in entropy, DS, during a transition between two equilibrium states requires finding a reversible path that connects the states. The entropy change calculated on that reversible path is taken to be DS for the actual path. This approach is necessary because quantities such as the temperature of a system can be defined only for systems in equilibrium, and a reversible path consists of a sequence of equilibrium states. The subscript r on the term Q_r emphasizes that the path chosen for the calculation must be reversible. The change in entropy ΔS, like changes in internal energy ΔU and changes in potential energy, depends only on the endpoints, and not on the path connecting them.

The concept of entropy gained wide acceptance in part because it provided another variable to describe the state of a system, along with pressure, volume, and temperature. Its significance was enhanced when it was found that **the entropy of the Universe increases in all natural processes**. This is yet another way of stating the second law of thermodynamics.

Although the entropy of the Universe increases in all natural processes, the entropy of a system can decrease. For example, if system A transfers energy Q to system B by heat, the entropy of system A decreases. This transfer, however, can only occur if the temperature of system B is less than the temperature of system A. Because temperature appears in the denominator in the definition of entropy, system B's increase in entropy will be greater than system A's decrease, so taken together, the entropy of the Universe increases.

For centuries, individuals have attempted to build perpetual motion machines that operate continuously without any input of energy or increase in entropy. The laws of thermodynamics preclude the invention of any such machines. The concept of entropy is satisfying because it enables us to present the second law of thermodynamics in the form of a mathematical statement. In the next section we find that entropy can also be interpreted in terms of probabilities, a relationship that has profound implications.

PROBLEM A block of ice at 273 K is put in thermal contact with a container of steam at 373 K, converting 25.0 g of ice to water at 273 K while condensing some of the steam to water at 373 K. (a) Find the change in entropy of the ice. (b) Find the change in entropy of the steam. (c) Find the change in entropy of the Universe.

STRATEGY First, calculate the energy transfer necessary to melt the ice. The amount of energy gained by the ice is lost by the steam. Compute the entropy change for each process and sum to get the entropy change of the Universe.

SOLUTION

(a) Find the change in entropy of the ice.

Use the latent heat of fusion, L_f, to compute the thermal energy needed to melt 25.0 g of ice:

$$Q_{\text{ice}} = mL_f = (0.025\ \text{kg})(3.33 \times 10^5\ \text{J}) = 8.33 \times 10^3\ \text{J}$$

Calculate the change in entropy of the ice:

$$\Delta S_{\text{ice}} = \frac{Q_{\text{ice}}}{T_{\text{ice}}} = \frac{8.33 \times 10^3\ \text{J}}{273\ \text{K}} = 30.5\ \text{J/K}$$

(b) Find the change in entropy of the steam.

By supposition, the thermal energy lost by the steam is equal to the thermal energy gained by the ice:

$$\Delta S_{\text{steam}} = \frac{Q_{\text{steam}}}{T_{\text{steam}}} = \frac{-8.33 \times 10^3\ \text{J}}{373\ \text{K}} = -22.3\ \text{J/K}$$

(c) Find the change in entropy of the Universe.

Sum the two changes in entropy:

$$\Delta S_{\text{universe}} = \Delta S_{\text{ice}} + \Delta S_{\text{steam}} = 30.5\ \text{J/k} - 22.3\ \text{J/K} = +8.2\ \text{J/K}$$

Entropy and Disorder

A large element of chance is inherent in natural processes. The spacing between trees in a natural forest, for example, is random; if you discovered a forest where all the trees were equally spaced, you would conclude that it had been planted. Likewise, leaves fall to the ground with random arrangements. It would be highly unlikely to find the leaves laid out in perfectly straight rows. We can express the results of such observations by saying that a disorderly arrangement is much more probable than an orderly one if the laws of nature are allowed to act without interference. Entropy originally found its place in thermodynamics, but its importance grew tremendously as the field of statistical mechanics developed. This analytical approach employs an alternate interpretation of entropy. In statistical mechanics, the behavior of a substance is described by the statistical behavior of the atoms and molecules contained in it. One of the main conclusions of the statistical mechanical approach is **that isolated systems tend toward greater disorder, and entropy is a measure of that disorder**.

In light of this new view of entropy, Boltzmann found another method for calculating entropy through use of the relation

$$S = k_B \ln W \qquad [2.15]$$

where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and W is a number proportional to the probability that the system has a particular configuration. The symbol "ln" again stands for natural logarithm, discussed in Appendix A.

Equation 2.15 could be applied to a bag of marbles. Imagine that you have 100 marbles—50 red and 50 green—stored in a bag. You are allowed to draw four marbles from the bag according to the following rules: Draw one marble, record its color, return it to the bag, and draw again. Continue this process until four marbles have been drawn. Note that because each marble is returned to the bag before the next one is drawn, the probability of drawing a red marble is always the same as the probability of drawing a green one.

The results of all possible drawing sequences are shown in Table 2.3. For example, the result RRGR means that a red marble was drawn first, a red one second, a green one third, and a red one fourth. The table indicates that there is only one possible way to draw four red marbles. There are four possible sequences that produce one green and three red marbles, six sequences that produce two green and two red, four sequences that produce three green and one red, and one sequence that

Table 2.3 Possible Results of Drawing Four Marbles from a Bag

End Result	Possible Draws	Total Number of Same Results
All R	RRRR	1
1G, 3R	RRRG, RRGR, RGRR, GRRR	4
2G, 2R	RRGG, RGRG, GRRG, RGGR, GRGR, GGRR	6
3G, 1R	GGGR, GGRG, GRGG, RGGG	4
All G	GGGG	1

produces all green. From Equation 2.15, we see that the state with the greatest disorder (two red and two green marbles) has the highest entropy because it is most probable. In contrast, the most ordered states (all red marbles and all green marbles) are least likely to occur and are states of lowest entropy.

The outcome of the draw can range between these highly ordered (lowest-entropy) and highly disordered (highest-entropy) states. Entropy can be regarded as an index of how far a system has progressed from an ordered to a disordered state.

The second law of thermodynamics is really a statement of what is most probable rather than of what must be. Imagine placing an ice cube in contact with a hot piece of pizza. There is nothing in nature that absolutely forbids the transfer of energy by heat from the ice to the much warmer pizza. Statistically, it's possible for a slow-moving molecule in the ice to collide with a faster-moving molecule in the pizza so that the slow one transfers some of its energy to the faster one. When the great number of molecules present in the ice and pizza are considered, however, the odds are overwhelmingly in favor of the transfer of energy from the faster moving molecules to the slower-moving molecules. Furthermore, this example demonstrates that a system naturally tends to move from a state of order to a state of disorder. The initial state, in which all the pizza molecules have high kinetic energy and all the ice molecules have lower kinetic energy, is much more ordered than the final state after energy transfer has taken place and the ice has melted.

Even more generally, the second law of thermodynamics defines the direction of time for all events as the direction in which the entropy of the universe increases. Although conservation of energy isn't violated if energy flows spontaneously from a cold object (the ice cube) to a hot object (the pizza slice), that event violates the second law because it represents a spontaneous increase in order. Of course, such an event also violates everyday experience. If the melting ice cube is filmed and the film speeded up, the difference between running the film in forward and reverse directions would be obvious to an audience. The same would be true of filming any event involving a large number of particles, such as a dish dropping to the floor and shattering.

As another example, suppose you were able to measure the velocities of all the air molecules in a room at some instant. It's very unlikely that you would find all molecules moving in the same direction with the same speed; that would be a highly ordered state, indeed. The most probable situation is a system of molecules moving haphazardly in all directions with a wide distribution of speeds, a highly disordered state. This physical situation can be compared to the drawing of marbles from a bag: If a container held 1023 molecules of a gas, the probability of finding all the molecules moving in the same direction with the same speed at some instant would be similar to that of drawing a marble from the bag 1023 times and getting a red marble on every draw, clearly an unlikely set of events.

The tendency of nature to move toward a state of disorder affects the ability of a system to do work. Consider a ball thrown toward a wall. The ball has kinetic energy, and its state is an ordered one, which means that all the atoms and molecules of the ball move in unison at the same speed and in the same direction (apart from their random internal motions). When the ball hits the wall, however, part of the ball's kinetic energy is transformed into the random, disordered, internal motion of the molecules in the ball and the wall, and the temperatures of the ball and the wall both increase slightly. Before the collision, the ball was capable of doing work. It could drive a nail into the wall, for example. With the transformation of part of the ordered energy into disordered internal energy, this capability

of doing work is reduced. The ball rebounds with less kinetic energy than it originally had, because the collision is inelastic.

Various forms of energy can be converted to internal energy, as in the collision between the ball and the wall, but the reverse transformation is never complete. In general, given two kinds of energy, A and B, if A can be completely converted to B and vice versa, we say that A and B are of the same grade. However, if A can be completely converted to B and the reverse is never complete, then A is of a higher grade of energy than B. In the case of a ball hitting a wall, the kinetic energy of the ball is of a higher grade than the internal energy contained in the ball and the wall after the collision. When high-grade energy is converted to internal energy, it can never be fully recovered as high-grade energy.

This conversion of high-grade energy to internal energy is referred to as the **degradation of energy**. The energy is said to be degraded because it takes on a form that is less useful for doing work. In other words, **in all real processes, the energy available for doing work decreases**.

Finally, note once again that the statement that entropy must increase in all natural processes is true only for isolated systems. There are instances in which the entropy of some system decreases, but with a corresponding net increase in entropy for some other system. When all systems are taken together to form the Universe, **the entropy of the Universe always increases**. Ultimately, the entropy of the Universe should reach a maximum. When it does, the Universe will be in a state of uniform temperature and density. All physical, chemical, and biological processes will cease, because a state of perfect disorder implies no available energy for doing work. This gloomy state of affairs is sometimes referred to as the ultimate “heat death” of the Universe.

PROBLEMS

1. An ideal gas is enclosed in a cylinder with a movable piston on top of it. The piston has a mass of 8 000 g and an area of 5.00 cm^2 and is free to slide up and down, keeping the pressure of the gas constant. (a) How much work is done on the gas as the temperature of 0.200 mol of the gas is raised from 20.0°C to 300°C? (b) What does the sign of your answer to part (a) indicate?
2. Sketch a PV diagram and find the work done by the gas during the following stages. (a) A gas is expanded from a volume of 1.0 L to 3.0 L at a constant pressure of 3.0 atm. (b) The gas is then cooled at constant volume until the pressure falls to 2.0 atm. (c) The gas is then compressed at a constant pressure of 2.0 atm from a volume of 3.0 L to 1.0 L. Note: Be careful of signs. (d) The gas is heated until its pressure increases from 2.0 atm to 3.0 atm at a constant volume. (e) Find the net work done during the complete cycle.
3. Gas in a container is at a pressure of 1.5 atm and a volume of 4.0 m^3. What is the work done on the gas (a) if it expands at constant pressure to twice its initial volume, and (b) if it is compressed at constant pressure to one-quarter its initial volume?

4. A 40.0-g projectile is launched by the expansion of hot gas in an arrangement shown in Figure P 2.4a. The cross-sectional area of the launch tube is 1.0 cm^2, and the length that the projectile travels down the tube after starting from rest is 32 cm. As the gas expands, the pressure varies as shown in Figure P2.4b. The values for the initial pressure and volume are $P_i = 11 \times 10^5$ Pa and $V_i \times 8.0$ cm^3 while the final values are $P_f = 1.0 \times 10^5$ Pa and $V_f \times 40.0$ cm^3. Friction between the projectile and the launch tube is negligible. (a) If the projectile is launched into a vacuum, what is the speed of the projectile as it leaves the launch tube? (b) If instead the projectile is launched into air at a pressure of 1.0×10^5 Pa, what fraction

of the work done by the expanding gas in the tube is spent by the projectile pushing air out of the way as it proceeds down the tube?

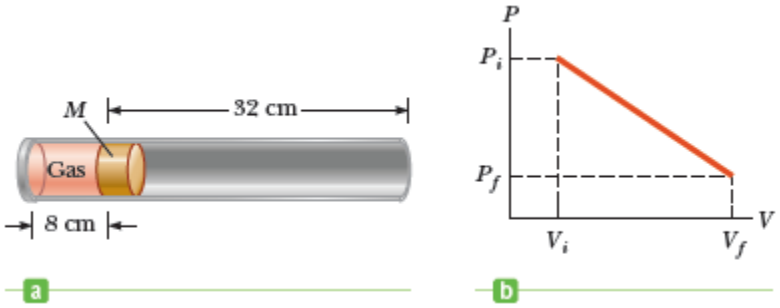


Figure P2.4

5. A gas expands from I to F along the three paths indicated in Figure P2.5. Calculate the work done on the gas along paths (a) IAF, (b) IF, and (c) IBF.

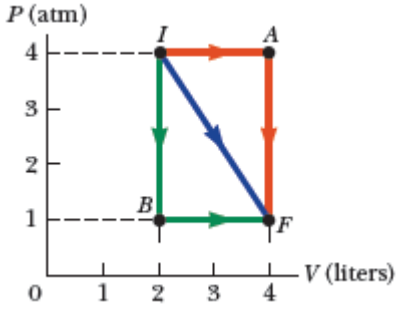


Figure P2.5 Problems 5 and 13.

6. Sketch a PV diagram of the following processes: (a) A gas expands at constant pressure P_1 from volume V_1 to volume V_2. It is then kept at constant volume while the pressure is reduced to P_2. (b) A gas is reduced in pressure from P_1 to P_2 while its volume is held constant at V_1. It is then expanded at constant pressure P_2 to a final volume V_2. (c) In which of the processes is more work done by the gas? Why?
7. A sample of helium behaves as an ideal gas as it is heated at constant pressure from 273 K to 373 K. If 20.0 J of work is done by the gas during this process, what is the mass of helium present?
8. (a) Find the work done *by* an ideal gas as it expands from point *A* to point *B* along the path shown in Figure P2.8. (b) How much work is done by the gas if it compressed from *B* to *A* along the same path?

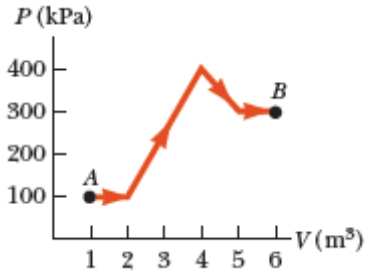


Figure P2.8

9. One mole of an ideal gas initially at a temperature of $T_i = 0°C$ undergoes an expansion at a constant pressure of 1.00 atm to four times its original volume. (a) Calculate the new temperature T_f of the gas. (b) Calculate the work done on the gas during the expansion.
10. (a) Determine the work done on a fluid that expands from *i* to *f* as indicated in Figure P2.10. (b) How much work is done on the fluid if it is compressed from *f* to *i* along the same path?

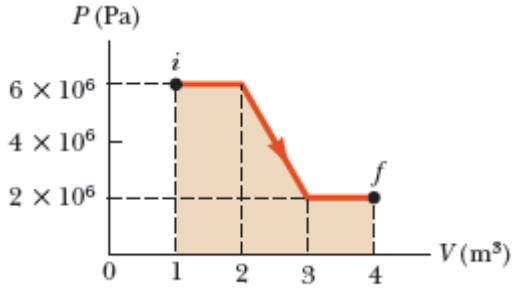


Figure P2.10

11. The only form of energy possessed by molecules of a monatomic ideal gas is translational kinetic energy. Using the results from the discussion of kinetic theory, show that the internal energy of a monatomic ideal gas at pressure P and occupying volume V may be written as $U = \frac{3}{2}PV$.
12. A cylinder of volume 0.300 m^3 contains 10.0 mol of neon gas at 20.0°C. Assume neon behaves as an ideal gas. (a) What is the pressure of the gas? (b) Find the internal energy of the gas. (c) Suppose the gas expands at constant pressure to a volume of 1.000 m^3. How much work is done on the gas? (d) What is the temperature of the gas at the new volume? (e) Find the internal energy of the gas when its volume is 1.000 m^3. (f) Compute the change in the internal energy during the expansion. (g) Compute ΔU - W. (h) Must thermal energy be transferred to the gas during the constant pressure expansion or be taken away? (i) Compute Q, the thermal energy transfer. (j) What symbolic relationship between Q, ΔU, and W is suggested by the values obtained?
13. A gas expands from *I* to *F* in Figure P2.5. The energy added to the gas by heat is 418 J when the gas goes from *I* to *F* along the diagonal path. (a) What is the change in internal energy of the gas? (b) How much energy must be added to the gas by heat for the indirect path *IAF* to give the same change in internal energy?
14. In a running event, a sprinter does 4.8 × 10^5 J of work and her internal energy decreases by 7.5 × 10^5 J. (a) Determine the heat transferred between her body and surroundings during this event. (b) What does the sign of your answer to part (a) indicate?
15. A gas is compressed at a constant pressure of 0.800 atm from 9.00 L to 2.00 L. In the process, 400 J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?
16. A quantity of a monatomic ideal gas undergoes a process in which both its pressure and volume are doubled as shown in Figure P2.16. What is the energy absorbed by heat into the gas during this process? Hint: The internal energy of a monatomic ideal gas at pressure P and occupying volume V is given by $U = \frac{3}{2}PV$

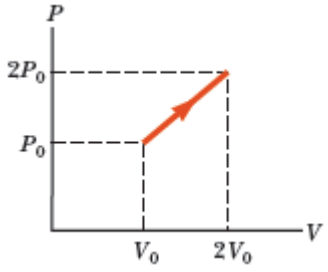


Figure P2.16

17. A gas is enclosed in a container fitted with a piston of cross-sectional area 0.150 m^2. The pressure of the gas is maintained at 6 000 Pa as the piston moves inward 20.0 cm. (a) Calculate the work done by the gas. (b) If the internal energy of the gas decreases by 8.00 J, find the amount of energy removed from the system by heat during the compression.
18. A monatomic ideal gas undergoes the thermodynamic process shown in the PV diagram of Figure P2.18. Determine whether each of the values ΔU, Q, and W for the gas is positive,

negative, or zero. Hint: The internal energy of a monatomic ideal gas at pressure P and occupying volume V is given by $U = \frac{3}{2}PV$

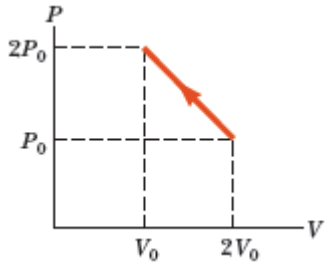


Figure P2.18

19. An ideal gas is compressed from a volume of V_i = 5.00 L to a volume of V_f = 3.00 L while in thermal contact with a heat reservoir at T= 295 K as in Figure P2.19. During the compression process, the piston moves down a distance of d = 0.130 m under the action of an average external force of F = 25.0 kN. Find (a) the work done on the gas, (b) the change in internal energy of the gas, and (c) the thermal energy exchanged between the gas and the reservoir. (d) If the gas is thermally insulated so no thermal energy could be exchanged, what would happen to the temperature of the gas during the compression?

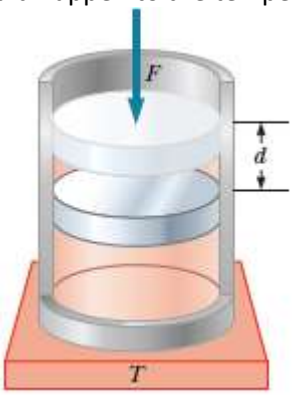


Figure P2.19

20. A system consisting of 0.025 6 moles of a diatomic ideal gas is taken from state A to state C along the path in Figure P2.20. (a) How much work is done on the gas during this process? (b) What is the lowest temperature of the gas during this process, and where does it occur? (c) Find the change in internal energy of the gas and (d) the energy delivered to the gas in going from A to C. Hint: For part (c), adapt the equation in the remarks of Example 2.9 to a diatomic ideal gas.

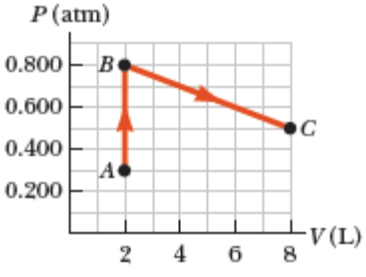


Figure P2.20

21. An ideal monatomic gas expands isothermally from 0.500 m^3 to 1.25 m^3 at a constant temperature of 675 K. If the initial pressure is 1.00×10^5 Pa, find (a) the work done on the gas, (b) the thermal energy transfer Q, and (c) the change in the internal energy.
22. An ideal gas expands at constant pressure. (a) Show that $P\Delta V = nR\Delta T$. (b) If the gas is monatomic, start from the definition of internal energy and show that $\Delta U = \frac{3}{2}W_{env}$, where W_{env} is the work done by the gas on its environment. (c) For the same monatomic ideal

gas, show with the first law that $Q = \frac{5}{2}W_{env}$. (d) Is it possible for an ideal gas to expand at constant pressure while exhausting thermal energy? Explain.

23. An ideal monatomic gas is contained in a vessel of constant volume 0.200 m^3. The initial temperature and pressure of the gas are 300 K and 5.00 atm, respectively. The goal of this problem is to find the temperature and pressure of the gas after 16.0 kJ of thermal energy is supplied to the gas. (a) Use the ideal gas law and initial conditions to calculate the number of moles of gas in the vessel. (b) Find the specific heat of the gas. (c) What is the work done by the gas during this process? (d) Use the first law of thermodynamics to find the change in internal energy of the gas. (e) Find the change in temperature of the gas. (f) Calculate the final temperature of the gas. (g) Use the ideal gas expression to find the final pressure of the gas.
24. Consider the cyclic process described by Figure P2.24. If Q is negative for the process BC and ΔU is negative for the process CA, determine the signs of Q, W, and ΔU associated with each process.

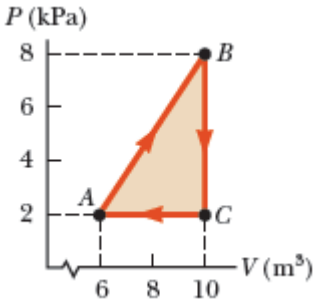


Figure P2.24

25. A 5.0-kg block of aluminum is heated from 20°C to 90°C at atmospheric pressure. Find (a) the work done by the aluminum, (b) the amount of energy transferred to it by heat, and (c) the increase in its internal energy.
26. One mole of gas initially at a pressure of 2.00 atm and a volume of 0.300 L has an internal energy equal to 91.0 J. In its final state, the gas is at a pressure of 1.50 atm and a volume of 0.800 L, and its internal energy equals 182 J. For the paths IAF, IBF, and IF in Figure P12.26, calculate (a) the work done on the gas and (b) the net energy transferred to the gas by heat in the process.

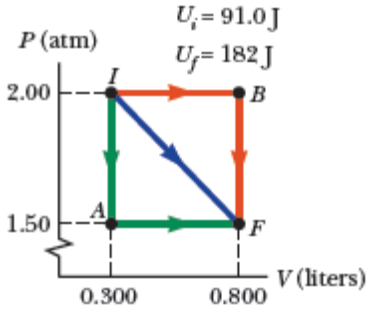


Figure P2.26

27. A gas increases in pressure from 2.00 atm to 6.00 atm at a constant volume of 1.00 m3 and then expands at constant pressure to a volume of 3.00 m3 before returning to its initial state as shown in Figure P 2.29. How much work is done in one cycle?

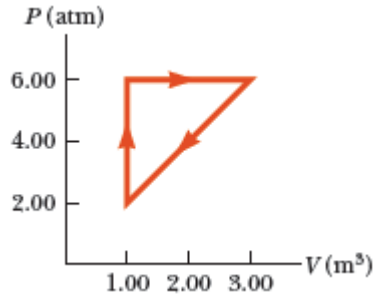


Figure P2.29

28. An ideal gas expands at a constant pressure of 6.00×10^5 Pa from a volume of 1.00 m^3 to a volume of 4.00 m^3 and then is compressed to one-third that pressure and a volume of 2.50 m^3 as shown in Figure P2.28 before returning to its initial state. How much work is done in taking a gas through one cycle of the process shown in the figure?

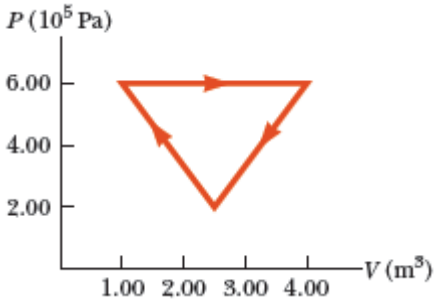


Figure P2.28

29. A heat engine operates between a reservoir at 25°C and one at 375°C. What is the maximum efficiency possible for this engine?
30. A heat engine is being designed to have a Carnot efficiency of 65% when operating between two heat reservoirs. (a) If the temperature of the cold reservoir is 20°C, what must be the temperature of the hot reservoir? (b) Can the actual efficiency of the engine be equal to 65%? Explain.
31. The work done by an engine equals one-fourth the energy it absorbs from a reservoir. (a) What is its thermal efficiency? (b) What fraction of the energy absorbed is expelled to the cold reservoir?
32. In each cycle of its operation, a heat engine expels 2 400 J of energy and performs 1 800 J of mechanical work. (a) How much thermal energy must be added to the engine in each cycle? (b) Find the thermal efficiency of the engine.
33. One of the most efficient engines ever built is a coal fired steam turbine engine in the Ohio River valley, driving an electric generator as it operates between 1 870°C and 430°C. (a) What is its maximum theoretical efficiency? (b) Its actual efficiency is 42.0%. How much mechanical power does the engine deliver if it absorbs 1.40 3 105 J of energy each second from the hot reservoir.
34. A gun is a heat engine. In particular, it is an internal combustion piston engine that does not operate in a cycle, but comes apart during its adiabatic expansion process. A certain gun consists of 1.80 kg of iron. It fires one 2.40-g bullet at 320 m/s with an energy efficiency of 1.10%. Assume the body of the gun absorbs all the energy exhaust and increases uniformly in temperature for a short time before it loses any energy by heat into the environment. Find its temperature increase.
35. A Styrofoam cup holding 125 g of hot water at 1.00×10^2°C cools to room temperature, 20.0°C. What is the change in entropy of the room? (Neglect the specific heat of the cup and any change in temperature of the room.)

36. A 65-g ice cube is initially at 0.0°C. (a) Find the change in entropy of the cube after it melts completely at 0.0°C. (b) What is the change in entropy of the environment in this process? Hint: The latent heat of fusion for water is 3.33×10^5 J/kg.
37. A freezer is used to freeze 1.0 L of water completely into ice. The water and the freezer remain at a constant temperature of T 5 0°C. Determine (a) the change in the entropy of the water and (b) the change in the entropy of the freezer.
38. What is the change in entropy of 1.00 kg of liquid water at 100°C as it changes to steam at 100°C?
39. A 70.0-kg log falls from a height of 25.0 m into a lake. If the log, the lake, and the air are all at 300 K, find the change in entropy of the Universe during this process.
40. If you roll a pair of dice, what is the total number of ways in which you can obtain (a) a 12? (b) a 7?
41. The surface of the Sun is approximately at 5 700 K, and the temperature of the Earth's surface is approximately 290 K. What entropy change occurs when 1 000 J of energy is transferred by heat from the Sun to the Earth?

